Proof

Let $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$, $\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$ and $\vec{c} = c_1\hat{i} + c_2\hat{j} + c_3\hat{k}$.

Then, $\vec{a}\cdot(\vec{b}\times\vec{c}) = (\vec{b}\times\vec{c})\cdot\vec{a} = \begin{vmatrix} b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \\ a_1 & a_2 & a_3 \end{vmatrix} = -\begin{vmatrix} a_1 & a_2 & a_3 \\ c_1 & c_2 & c_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$, by $R_1 \leftrightarrow R_3$

$= \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$, by $R_2 \leftrightarrow R_3$

$= (\vec{a}\times\vec{b})\cdot\vec{c}$.

Hence the theorem is proved. ■

Note

By Theorem 6.2, it follows that, in a scalar triple product, dot and cross can be **interchanged without altering the order of occurrences of the vectors**, by placing the parentheses in such a way that dot lies outside the parentheses, and cross lies between the vectors inside the parentheses. For instance, we have

$(\vec{a}\times\vec{b})\cdot\vec{c} = \vec{a}\cdot(\vec{b}\times\vec{c})$, since dot and cross can be interchanged.

$= (\vec{b}\times\vec{c})\cdot\vec{a}$, since dot product is commutative.

$= \vec{b}\cdot(\vec{c}\times\vec{a})$, since dot and cross can be interchanged

$= (\vec{c}\times\vec{a})\cdot\vec{b}$, since dot product is commutative

$= \vec{c}\cdot(\vec{a}\times\vec{b})$, since dot and cross can be interchanged

Notation

For any three vectors $\vec{a}, \vec{b}$ and $\vec{c}$, the scalar triple product $(\vec{a}\times\vec{b})\cdot\vec{c}$ is denoted by $[\vec{a}, \vec{b}, \vec{c}]$.

$[\vec{a}, \vec{b}, \vec{c}]$ is read as **box $\vec{a}, \vec{b}, \vec{c}$**. For this reason and also because the absolute value of a scalar triple product represents the volume of a box (rectangular parallelepiped),a scalar triple product is also called a **box product**.

Note

(1) $[\vec{a}, \vec{b}, \vec{c}] = (\vec{a}\times\vec{b})\cdot\vec{c} = \vec{a}\cdot(\vec{b}\times\vec{c}) = (\vec{b}\times\vec{c})\cdot\vec{a} = \vec{b}\cdot(\vec{c}\times\vec{a}) = [\vec{b}, \vec{c}, \vec{a}]$

$[\vec{b}, \vec{c}, \vec{a}] = (\vec{b}\times\vec{c})\cdot\vec{a} = \vec{b}\cdot(\vec{c}\times\vec{a}) = (\vec{c}\times\vec{a})\cdot\vec{b} = \vec{c}\cdot(\vec{a}\times\vec{b}) = [\vec{c}, \vec{a}, \vec{b}]$.

In other words, $[\vec{a}, \vec{b}, \vec{c}] = [\vec{b}, \vec{c}, \vec{a}] = [\vec{c}, \vec{a}, \vec{b}]$; that is, if the three vectors are permuted in the same cyclic order, the value of the scalar triple product remains the same.

(2) If any two vectors are interchanged in their position in a scalar triple product, then the value of the scalar triple product is (-1) times the original value. More explicitly,

$[\vec{a}, \vec{b}, \vec{c}] = [\vec{b}, \vec{c}, \vec{a}] = [\vec{c}, \vec{a}, \vec{b}] = -[\vec{a}, \vec{c}, \vec{b}] = -[\vec{c}, \vec{b}, \vec{a}] = -[\vec{b}, \vec{a}, \vec{c}]$.

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